

A Modular General-Purpose Semi-Humanoid Robotic Platform

Technical Project Overview and Conceptual Mechatronic Architecture

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Abstract

This report presents the conceptual design, systems engineering architecture, and technical specification for an 18-Degree-of-Freedom (DoF) modular semi-humanoid robotic platform. Grounded in the VDI 2206 mechatronic systems design methodology, the platform addresses the trilemma between single-purpose automated machinery and cost-prohibitive bipedal humanoids. The robot integrates a high-traction differential-drive wheeled base, a fixed ergonomic space-frame torso, dual 6-DoF anthropomorphic arms featuring a hybrid Quasi-Direct Drive (QDD) and serial bus servo actuation topology, symmetrical parallel-jaw grippers, an active 2-DoF RGB-D perception head, and an isolated 24V LiFePO₄ power infrastructure. To decouple physical execution from cognitive intent, the system formalizes a sandboxed *Task Port* and a 4-tier multi-rate software hierarchy (5 Hz AI cognition to 1 kHz hard real-time motor commutation). Validated against standardized RoboCup@Home benchmarks, the core platform achieves an operational reach of 500–1350 mm, an all-up mass ≤ 55 kg, and an itemized build budget of 432,915 EGP ($\sim \$8,900$ USD).

Keywords: Semi-Humanoid Robot; VDI 2206 Mechatronics; Task Port; Hybrid QDD Actuation; Multi-Rate Architecture.

1 Executive Context, Motivation, and Design Philosophy

The Service Robotics Trilemma: Service, domestic, and laboratory automation currently faces a sharp technological divide. Rigid single-purpose robotic workcells lack task adaptability, while full-scale bipedal humanoids incur prohibitive capital costs ($> \$100k$ – $\$250k$), severe balance instabilities, and non-negligible idle energy consumption. As established in contemporary Cyber-Physical Systems (CPS) literature for Industry 5.0, a wheeled *semi-humanoid* mobile manipulator resolves this trilemma: a differential-drive base ensures unconditional static overturning stability, while an anthropomorphic upper body provides manipulation dexterity in human-designed spaces.

Core Architectural Principle – Capability vs. Intent: Traditional mobile robots hardcode task-specific behaviors into low-level control loops. In this project, we enforce a strict separation of concerns:

“The platform provides deterministic physical capability; the application layer provides cognitive intent.”

The physical body exposes a fixed, stable set of verified motion and perception primitives via a sandboxed **Task Port**. Downstream users, task-level Behavior Trees, or foundation Vision-Language-Action (VLA) models interact exclusively through standardized ROS 2 Action interfaces, preventing algorithmic software failures from compromising hardware safety.

VDI 2206 Mechatronic Methodology: System development strictly follows the VDI 2206 guideline [1]. The macro V-model guides progression from empirical task analysis to cross-domain conceptual design, followed by concurrent mechanical, electrical, and software engineering, ascending through Software-in-the-Loop (SIL), Hardware-in-the-Loop (HIL), and physical verification. Micro-cycles of problem-solving (*Situation Analysis* $\rightarrow$ *Target Formulation* $\rightarrow$ *Synthesis* $\rightarrow$ *Analysis/Evaluation* $\rightarrow$ *Decision* $\rightarrow$ *Planning*) prevent premature hardware locking.

In machine dynamics, joint stiffness K and distal moving mass M govern the mechanical natural frequency $\omega_n = \sqrt{K/M}$ and closed-loop settling time $t_{\text{settling}} \approx 4/(\zeta\omega_n)$. Minimizing distal link inertia by concentrating mass proximally raises structural resonance, suppresses vibrations, and enables high-gain, compliant manipulation.

2 Empirical Task Formulation, Benchmarks, and Requirements

Empirical Benchmarks: Rather than designing around an arbitrary task, physical boundaries are derived from international service robotics standards (RoboCup@Home [2], World Robot Summit, and ISO 13482:2014 safety standards [3]):

- *Doorway & Aisle Clearance:* Standard interior apertures (750–900 mm) dictate base width ≤ 550 mm for collision-free passage.
- *Working Height Envelopes:* Standard dining tables (750 mm), kitchen counters (900 mm), and mid-level shelving (1200 mm) establish a continuous required vertical reach band of 500–1350 mm above floor level.
- *Everyday Workpiece Masses:* Domestic tableware, beverage containers, and packaged items (0.1–1.0 kg) require ≥ 1.0 kg nominal continuous payload capacity at full horizontal arm extension (625 mm).

Platform Validation Tasks: The integrated platform is validated against two canonical reference benchmarks:

1. *Primary Benchmark (Autonomous Voice Pick-and-Place):* Speech input parsed by Whisper STT [10]; intent resolved by an OpenVLA [6] planner into typed Task Port actions; differential base navigates to table via Nav2 [9]; RealSense head estimates 6D object pose; MoveIt 2 [8] executes collision-free grasp; base transits and deposits item onto target surface.

2. *Secondary Benchmark (Web Dashboard Teleoperation & Task Programming)*: Operator jogs base and arms via low-latency WebRTC browser joysticks, monitors real-time telemetry/alarms, and records waypoint macros for autonomous trajectory replay.

Engineering Requirements Specification (*Anforderungsliste*): Following VDI 2206, Table 1 compiles the foundational engineering requirements, classified into Fixed Requirements (FF / Demand) and Range Requirements (BF / Wish), quantitative (*quan*) and qualitative (*qual*).

Table 1: Consolidated VDI 2206 Engineering Requirements Specification (*Anforderungsliste*).

ID	Category	Parameter Metric	Target Value	Type	Verification Method
G01	Geometry	Footprint envelope diameter	≤ 550 mm (Length ≤ 600 mm)	BF / quan	CAD audit & physical passage test
G02	Geometry	Doorway aperture passage	Clearance margin ≥ 100 mm in 750 mm doors	FF / quan	Physical benchmark doorway trial
G03	Geometry	Platform height & total mass	Height: 1200–1300 mm; Wet mass ≤ 55 kg	BF / quan	Coordinate gauge & calibrated scale
K01	Kinematics	Active Degrees of Freedom (DoF)	18 Active DoF (Base: 2, Arms: 12, Grippers: 2, Head: 2)	FF / quan	Topological kinematic joint audit
K02	Kinematics	Base drivetrain topology	Coaxial differential drive + dual swivel casters	FF / qual	Trajectory tracking & pivot-spin trial
K03	Kinematics	Vertical manipulation envelope	500 mm (table-low) to ≥ 1350 mm (high shelf)	FF / quan	TCP reach calibration across workspace
K04	Kinematics	Horizontal dual-arm bimanual span	Symmetric span ≥ 1200 mm; single arm reach ≥ 625 mm	BF / quan	Forward kinematic envelope mapping
D01	Dynamics	Arm payload capacity at full reach	Nominal: ≥ 1.0 kg; Peak static holding: ≥ 1.5 kg	FF / quan	Static holding test at full horizontal reach
D02	Dynamics	Parallel gripper stroke & clamping	Stroke: 0–85 mm; Clamping force: 15–35 N	FF / quan	Calibrated load-cell grip trial
D03	Dynamics	Base transit speed	0.6–0.8 m/s (ISO 13482 indoor safety limit)	BF / quan	Laser-timed linear velocity profile
D04	Dynamics	Cartesian repeatability	$\leq \pm 2.0$ mm under full payload	BF / quan	Dial indicator multi-cycle test
D05	Dynamics	Structural settling time	$t_{\text{settling}} \leq 0.5$ s after rapid slew ($\zeta \geq 0.6$)	BF / quan	Accelerometer vibration decay probe
E01	Electrical	Primary DC power bus	24.0 V nominal (8S LiFePO ₄ battery pack)	FF / quan	Oscilloscope rail voltage logging
E02	Electrical	Operational battery runtime	≥ 3.0 h continuous mixed cycle (≥ 5.0 h idle)	BF / quan	Automated execution loop to low-SoC
E03	Electrical	Hardware E-Stop isolation	Actuator power disconnected in ≤ 10 ms	FF / quan	Oscilloscope timing of contactor drop
C01	Control	Joint control cycle frequency	≥ 500 Hz (Target: 1000 Hz) on native MCU	FF / quan	FreeRTOS task jitter telemetry
C02	Control	Subsystem communication fieldbus	CAN 2.0B / TWAI (≥ 1 Mbps) & half-duplex TTL	FF / qual	Protocol analyzer packet loss check
C03	Control	Device-level modular abstraction	Each assembly operates as an independent node	FF / qual	Hot-unplug diagnostic isolation audit
S01	Sensing	Base obstacle detection coverage	360° planar LiDAR + ultrasonic blind-spot ring	FF / quan	Blind-spot mapping with standard targets
S02	Sensing	Active visual perception	Active stereo RGB-D camera on 2-DoF gimbal	FF / qual	Point cloud density & registration audit
A01	Software	Middleware framework	ROS 2 LTS (Humble/Jazzy) [7] with PREEMPT_RT	FF / qual	DDS QoS latency & jitter audit
A02	Software	Task Port abstraction interface	Sandboxed Action Servers (‘/navigate’, ‘/pick’, ‘/place’)	FF / qual	End-to-end task script invocation
H01	HMI	Web-based operational dashboard	React/WebRTC dashboard with teleop latency ≤ 80 ms	FF / quan	Network packet round-trip time probe
H02	HMI	Physical torso maintenance panel	OLED status display, RJ-45 GbE & dual USB 3.0 ports	FF / qual	Functional inspection of bulkhead panel
SF1	Safety	Human-co-presence safety	Dynamic speed scaling & force veto per ISO 13482	FF / qual	Contact stop & obstacle detection tests

Rationale for Core Architectural Decisions:

1. *Differential vs. Mecanum Drive*: Mecanum wheels exhibit 15–25% transverse dead-reckoning slip on smooth/carpeted floors, induce severe high-frequency roller vibrations that degrade camera point clouds, and waste battery power. Differential drive provides high traction, simple odometry, zero-radius pivot spinning, and rugged reliability.
2. *Dual 6-DoF Arms over 7-DoF*: 6-DoF arms fully span $SE(3)$ task space. Omitting a redundant 7th joint eliminates 1.5–2.0 kg of cantilever mass per arm, raising the joint natural frequency ω_n , reducing settling time, eliminating kinematic singularity search latency, and halving control bus packet overhead.

3. *Fixed Torso over Telescopic Column*: Motorized vertical lift columns add > 12 kg of dead mass, elevate the center of gravity, and suffer mechanical guide binding under bimanual cantilever loads. A fixed structural mast (shoulder axis at 950 mm) spans 500–1350 mm reach, covering tables, counters, and eye-level shelves at zero mechanical risk.
4. *2-Finger Parallel Grippers over Multi-Finger Hands*: Underactuated multi-finger hands suffer frequent cable stretch and low pinch forces. Symmetrical parallel-jaw grippers with Shore-A 60 compliant silicone pads deliver deterministic force/form closure (15–35 N) and sensorless grasp verification via motor phase current monitoring.

3 Functional Flow Modeling and System Mechatronic Architecture

Functional Structure (VDI 2206): The robot is modeled as a cybernetic system transforming three physical flows across its boundary $\mathcal{B}_{\text{system}}$: Material (**M**), Energy (**E**), and Information (**I**), as illustrated in Fig. 1.

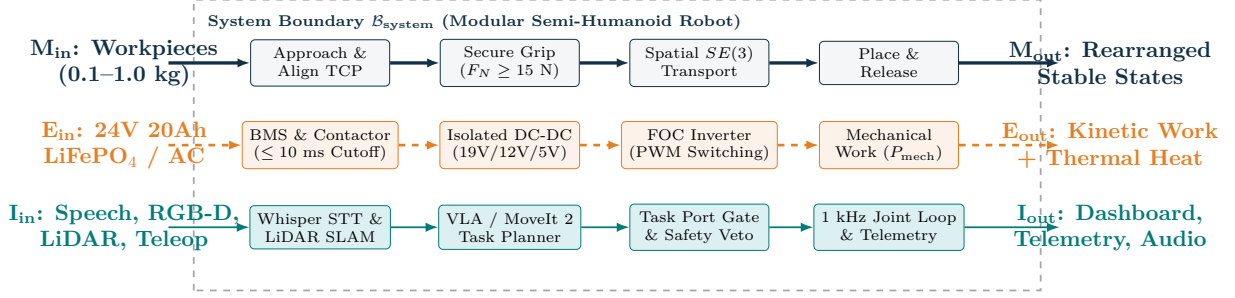


Figure 1: Tripartite functional flow structure decomposition (**M**, **E**, **I**) across the mechatronic system boundary.

Hierarchical Multi-Rate Execution Topology: To maintain control stability, software tasks are partitioned across four explicit frequency domains:

- *Tier 1: Cognition & HRI (5–10 Hz)*: Whisper STT, OpenVLA action chunk proposal, WebRTC video streaming.
- *Tier 2: Mission & Planning (30–50 Hz)*: BehaviorTree.CPP task sequencer, Nav2 local costmaps, MoveIt 2 Cartesian trajectory planning.
- *Tier 3: Robot Integration & Middleware (100–250 Hz)*: ‘ros2_control’ resource management, cubic Hermite spline interpolation, Unscented Kalman Filter (UKF) state estimation.
- *Tier 4: Safety & Actuation (500–1000 Hz)*: Native FreeRTOS loops on distributed ESP32 MCUs, 1 Mbps CAN bus cyclic commutation, actuator FOC current loops, thermal watchdogs, and hardware E-Stop monitoring.

4 Detailed Subsystem Engineering (The 8 Modular Devices)

In accordance with the *Device-Level* architecture, each subsystem is encapsulated as an independent node coordinated by an ESP32 microcontroller over CAN (TWAI) or USB/Ethernet to the primary NVIDIA Jetson Orin NX host.

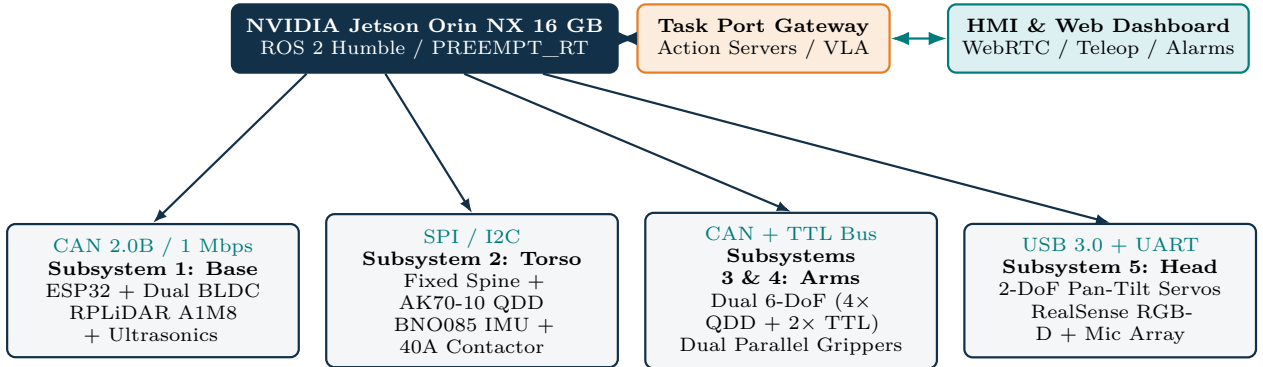


Figure 2: System mechatronic topology: Tier-1 real-time fieldbuses, central edge compute host, and Task Port gateway.

1. Subsystem 1: Mobile Base Device: Differential drive kinematics powered by $2 \times 24\text{V } 100\text{W}$ BLDC gearmotors ($\sim 30 : 1$ planetary reduction, optical quadrature encoders) driving 200 mm vulcanized rubber wheels. Dual 100 mm swivel casters provide a stable 4-point footprint clearing 15 mm cable thresholds. Perception includes a Slamtec RPLiDAR A1M8 (360° planar scan, 12 m range) fused with $4 \times \text{HC-SR04}$ ultrasonic sensors for blind-zone detection. Controlled by an ESP32-WROOM-32 via a custom dual H-bridge carrier PCB.

2. Subsystem 2: Fixed Torso & Structural Mast: A rigid 6061-T6 aluminum extrusion spine (30×30 mm and 40×40 mm) with shoulder yoke at 950 mm. Incorporates a waist tilt joint powered by a CubeMars AK70-10

QDD actuator (24.8 N · m peak torque, 10:1 planetary reduction, 14-bit absolute encoder, integrated FOC driver) and a BNO085 9-DoF IMU. Houses the main power contactor, fuse blocks, and active compute cooling.

3. Subsystems 3 & 4: Dual 6-DoF Manipulator Arm Devices: Anthropomorphic 6-DoF configuration ($SE(3)$ reach: 625 mm) featuring an innovative hybrid actuation topology:

- *Proximal Joints (Shoulder Pitch/Roll/Yaw, Elbow Pitch):* 4× CubeMars AK70-10 QDD actuators per arm, delivering low gear ratios (10 : 1), exceptional backdrivability, high impact resistance, and proprioceptive current-based torque sensing.
- *Distal Wrist Joints (Wrist Pitch and Roll):* 2× Feetech STS3215 serial bus servos (12V, 2.9 N · m stall, half-duplex TTL UART @ 1 Mbps), reducing distal arm weight by > 90% and actuator cost by > 75%.

Coordinated by a dedicated ESP32 per arm on a custom interface PCB bridging the 3.3V CAN transceiver to the TTL servo chain.

4. End-Effector: Dual 2-Finger Parallel Grippers: Driven by a Feetech STS3032 compact bus servo via a dual-rack-and-pinion transmission (0–85 mm stroke). Replaceable Shore-A 60 compliant silicone gripping pads ($\mu_s \geq 0.85$) conform passively to varied geometries. Grasp force is regulated from 15–35 N via closed-loop servo current feedback.

5. Subsystem 5: Active Perception Head & Neck Device: 2-DoF pan-tilt gimbal (2× Feetech STS3215 servos, 180°/s slew rate) carrying an Intel RealSense RGB-D depth camera (USB 3.0) and a ReSpeaker 2-Mic array for Acoustic Echo Cancellation (AEC) and Time Difference of Arrival (TDoA) sound localization, paired with a 2W front speaker.

6. Subsystem 6: Power Distribution & Safety Subsystem: 24V nominal, 20Ah (480 Wh) LiFePO₄ battery pack with smart BMS. A 24V 40A continuous safety contactor cuts motor rail power in ≤ 10 ms upon tripping the latching torso mushroom E-Stop switch. Synchronous buck regulators deliver isolated, noise-filtered 19V (Jetson), 12V (servos/sensors), and 5V (MCUs) logic rails, ensuring compute and logging survive emergency stops.

7. Subsystem 7: Central Compute & Communications: NVIDIA Jetson Orin NX 16 GB (100 TOPS AI compute) running PREEMPT_RT Linux, pinned across dedicated CPU cores ('ros2_control' on Cores 0–1; Nav2/MoveIt on Cores 2–5; Web services on Cores 6–7). Isolated USB-to-CAN gateway, Gigabit Ethernet switch, and dual-band Wi-Fi 6.

8. Subsystem 8: HMI, Web Dashboard & Task Port Gateway:

- *Physical Torso Panel:* Bulkhead panel with 5–7" OLED status display (IP, battery SoC, node states, error codes), RJ-45 GbE port, dual USB 3.0 service ports, and system reset button.
- *Browser Web Dashboard:* Single-page application (React, WebRTC, rosbridge) streaming < 80 ms video, 3D LiDAR occupancy maps, virtual joystick teleop, visual/audible alarms, and macro waypoint trajectory recording.
- *Task Port Gateway:* Exposes decoupled ROS 2 Actions (/navigate_to_pose, /pick_object, /place_object, /dock_robot), enforcing the capability boundary.

5 Domain Allocation, Bill of Materials, Roadmap, and Conclusion

Multidisciplinary Domain Allocation (VDI 2206): The mechatronic system design allocates responsibilities across three concurrent engineering domains (Table 2).

Table 2: Cross-Domain Engineering Work Breakdown Structure.

Domain	Primary Responsibilities & Work Packages	Key Verification Deliverables
Mechanical	Base chassis FEA, structural mast	CAD, static tipping margin ≥ 1.8 , settling time $t_{\text{settling}} \leq 0.5$ s,
	yoke, 6-DoF arm linkages, parallel grippers	ISO 9409-1 tool flanges, total dry mass ≤ 45 kg.
Electrical	Power PCB (BMS, 40A contactor, DC-DC bucks),	Gerber packages, ≤ 10 ms E-Stop cutoff, brownout-free
	Arm/Base/Head carrier PCBs, CAN bus cabling	rail transients, ≥ 3.0 h battery runtime under load.
Software	FreeRTOS MCU firmware (CAN/TTL), 'ros2_control',	Deterministic 1 kHz joint loop, Nav2 docking < 30 mm,
	Nav2, MoveIt 2, Task Port, Web Dashboard	MoveIt IK < 5 ms, WebRTC teleop latency ≤ 80 ms.

Bill of Materials & Economics: The prototype build is fully sourced using Egyptian market procurement data from *Budget Estimationn.pdf*. Table 3 outlines the cost share across subsystems.

Economic Sizing Rationale: Actuation comprises 52.0% of the platform budget, Compute & Control 27.3%, Perception 15.2%, and Power/Structure 5.5%. The hybrid actuation approach (CubeMars QDDs on high-torque proximal joints + Feetech bus servos on distal joints) delivers a > 70% cost saving over commercial collaborative arms (e.g., Franka Emika or UR3e), rendering the platform reproducible for academic research.

Implementation Roadmap & Risk Mitigation: The project spans 4 synchronized phases across 9 months:

- *Phase 1 (Months 1–3): Component Benchtop Testing & Custom PCB Prototyping.* Dynamic dynamometer profiling of AK70-10 QDDs, fabrication of Base/Arm/Power carrier PCBs, and 1 Mbps CAN bus timing validation.

Table 3: Consolidated Platform Bill of Materials (BOM) Cost Breakdown.

Subsystem Assembly	Cost (EGP)	Share (%)	Key Components Included
Subsystem 1: Mobile Base	31,590	7.3%	Dual BLDC hub motors, 200 mm rubber wheels, casters, RPLiDAR A1M8, ESP32, Base PCB.
Subsystems 2 & 6: Torso & Power	148,160	34.2%	AK70-10 QDD, Jetson Orin NX 16 GB, 24V 20Ah LiFePO ₄ battery, 40A contactor, Power PCB.
Subsystem 5: Perception Head	62,615	14.5%	2× Feetech STS3215 servos, Intel RealSense RGB-D, ReSpeaker Mic, speaker, ESP32.
Subsystems 3 & 4: Dual Arms	190,550	44.0%	8× AK70-10 QDDs, 4× STS3215 servos, 2× STS3032 grippers, 2× ESP32, Arm PCBs.
CORE PLATFORM TOTAL	EGP 432,915	100.0%	Complete 18-DoF Semi-Humanoid Robotic Platform (~\$8,900 USD).
<i>Modular Telescopic Column (Opt.)</i>	<i>+32,920</i>	<i>+7.6%</i>	<i>24V linear actuator, 300 mm stroke, 1500 N thrust, guide rails, driver PCB.</i>

- *Phase 2 (Months 3–5): SIL Simulation & Digital Twin Development.* High-fidelity URDF/USD simulation in NVIDIA Isaac Sim and Gazebo; verification of MoveIt 2 bimanual trajectory tracking and Nav2 costmaps.
- *Phase 3 (Months 5–7): Physical Integration & Hardware-in-the-Loop (HIL).* Mechanical frame assembly, wiring harness routing, safety contactor E-Stop drop tests, and thermal heat soak validation.
- *Phase 4 (Months 7–9): Task Port Validation & Proof-of-Concept Benchmarking.* Full deployment of the primary voice-commanded pick-and-place pipeline and web teleoperation console on RoboCup@Home mock scenarios.

Conclusion & Research Outlook: This project establishes the conceptual mechatronic foundation for a modular, general-purpose semi-humanoid robotic platform. By formalizing the Task Port abstraction boundary and combining a high-traction differential base with a fixed-mast hybrid QDD upper body, the platform bridges the gap between brittle industrial automation and unconstrained AI cognitive agents. Future research will explore Whole-Body Operational Space Control (WBC) dynamically coordinating base and arms, and domain-randomized sim-to-real policy transfer.

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